Weighted Automata Learning with a Hard Inductive Bias

Breandan Considine

Main Idea

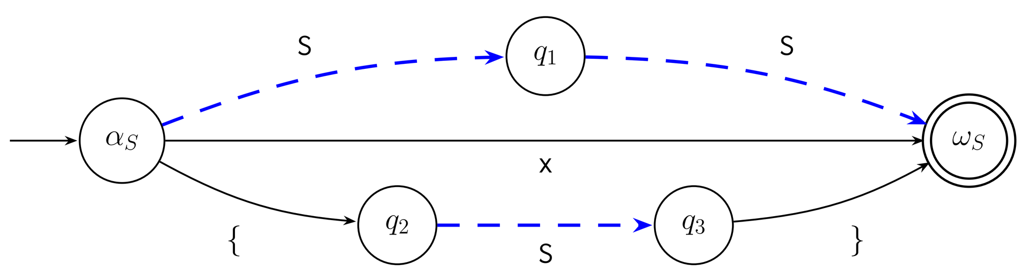
- **Assume**: we are given a learned distribution, $P_\theta(\sigma)$, with support over $\sigma : \Sigma^+$ but largely concentrated within $\mathcal{L}(G)$, where G is a known CFG.
- **Our goal**: distill a surrogate WFA, $\alpha_{\hat\theta}$, such that for all $\sigma \in \mathcal{L}(G)$, $P_{\hat\theta}(\sigma) \propto P_\theta\big(\sigma \mid \sigma \in \mathcal{L}(G)\big)$, while preserving inclusion $\big(\mathcal{L}(\alpha) \supseteq \mathcal{L}(G)\big)$.
- **Desiderata**: The regular description α should be reasonably concise and precise, i.e., $\Delta = |\mathcal{L}(\alpha) \setminus \mathcal{L}(G)|$ is approximately minimized up to Σ^n.

Nederhof's Construction

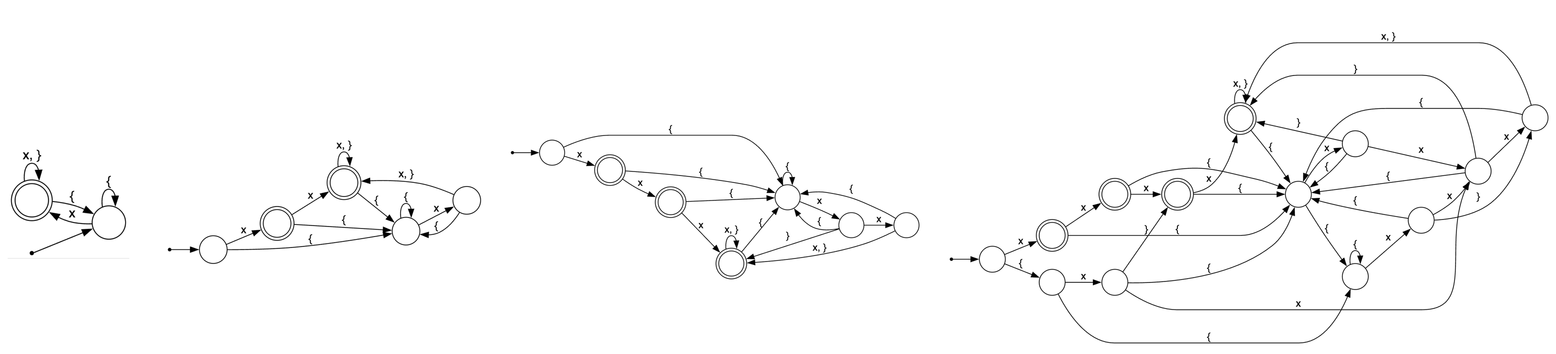
A recursive transition network (RTN) of a CFG $\langle \Sigma, V, P, S\rangle$ is an automaton,

$$\mathcal{R} = \big\{R_A = (Q_A, \delta_A, \alpha_A, F_A)\big\}_{A\in V}, \quad \delta_A \subseteq Q_A \times \big(\Sigma \cup V \cup \{\varepsilon\}\big) \times Q_A,$$

where an arc labeled $B \in V$ denotes a call to R_B. For example, consider the CFG, `S → x | S S | { S }`, which has the following RTN:

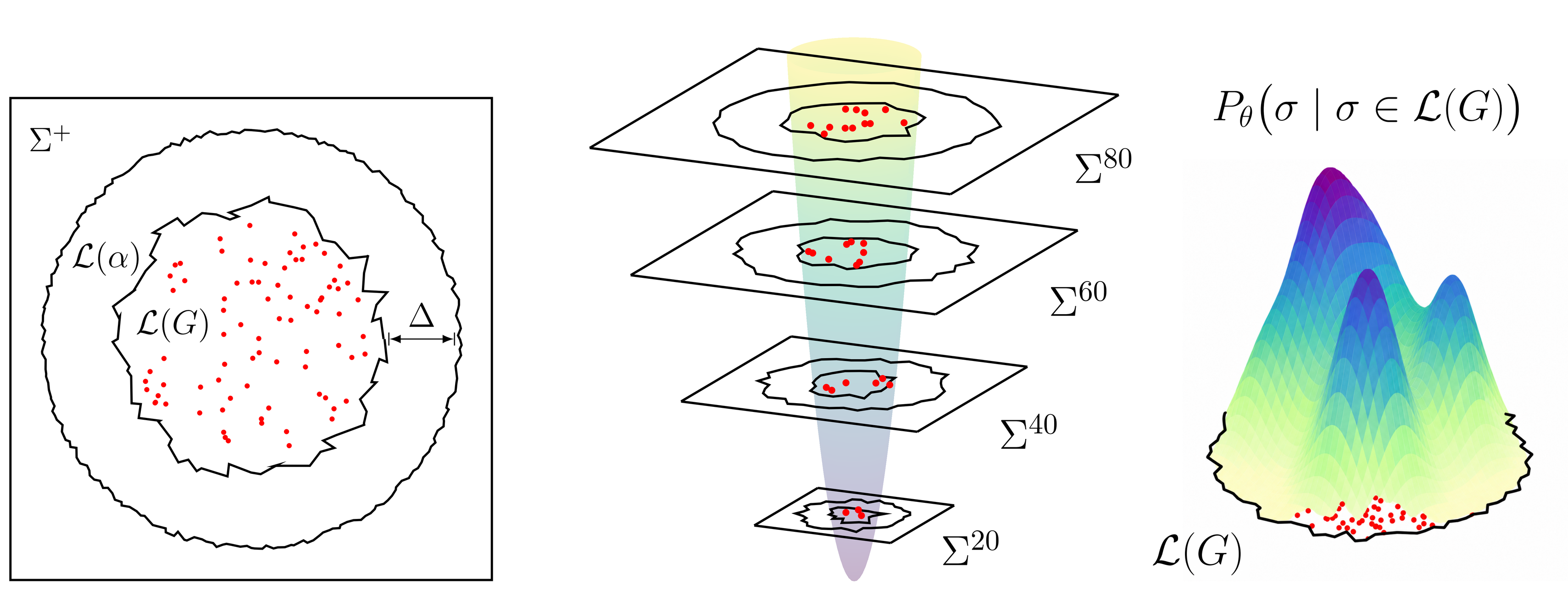


Nederhof's d-RTN approximation is a depth-bounded RTN unfolding. A nonterminal arc $p \xrightarrow{B} q$ is replaced by a fresh copy of the RTN for B, adding ε-arcs from p to the copy's initial state and the copy's final state back to q: $p \xrightarrow{\varepsilon} \alpha'_B,\ \ \omega'_B \xrightarrow{\varepsilon} q \quad (\forall \omega'_B \in F'_B)$. Unfold calls for d rounds. At depth d, every remaining nonterminal arc $p \xrightarrow{B} q$ is woven into a shared copy of R_B: $p \xrightarrow{\varepsilon} \alpha_B,\ \ \omega_B \xrightarrow{\varepsilon} q \quad (\forall \omega_B \in F_B)$. Shown below is the ε-removed, determinized and minimized DFA rendered for $1 \leq d \leq 4$:



$d = 1$ $\quad$ $d = 2$ $\quad$ $d = 3$ $\quad$ $d = 4$

Language Overapproximation



We can measure over-approximation tightness by considering finite models,

$$\rho(\alpha, G, m) = \frac{\sum_{n=1}^{m} |\mathcal{L}(G) \cap \Sigma^n|}{\sum_{n=1}^{m} |\mathcal{L}(\alpha) \cap \Sigma^n|}, \alpha \sqsubset_G^m \alpha' \iff \rho(\alpha', G, m) < \rho(\alpha, G, m).$$

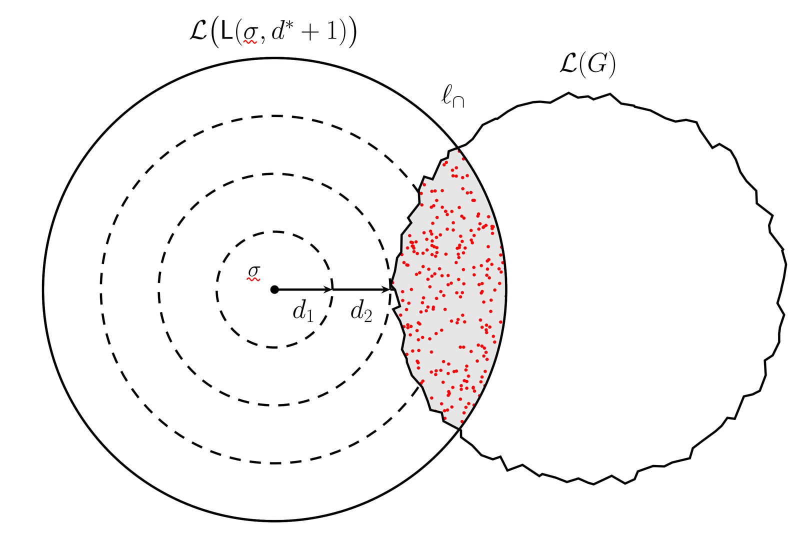
Method

We propose a counterexample-guided abstraction refinement (CEGAR) loop that incrementally refines $\alpha_{\hat\theta}$ until a desired precision or $|Q|_{\max}$ is reached:

1. Initialize $\alpha'_{\hat\theta} \leftarrow \alpha_d$ using Nederhof's d-RTN construction.
2. Propose a refinement $D_{\mathsf{KL}}(P_\theta \| \alpha_{\hat\theta}) < D_{\mathsf{KL}}(P_\theta \| \alpha'_{\hat\theta})$ and $\alpha_{\hat\theta} \sqsubset_G^m \alpha'_{\hat\theta}$.
3. Binarize $\alpha_{\hat\theta}$ and verify $\mathcal{L}(\alpha) \supset \mathcal{L}(G)$ by checking $\varnothing = \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$.
4. If verification fails, collect $x \in \mathcal{L}(G) \cap \overline{\mathcal{L}(\alpha)}$, otherwise go to step 2.
5. Repair α restoring inclusion, renormalize as $\alpha'_{\hat\theta}$, then go to step 2.

Intersection Decoding

We will evaluate our learned WFA on a constrained decoding task: given a dataset of near-words $\underset{\sim}{\sigma} \notin \mathcal{L}(G)$ and pairwise human repairs σ' within a small edit distance, sample uniformly WoR repairs within $\ell_\cap$, score and rank each repair according to $P_{\hat\theta}(\sigma)$, then measure the true repair's rank.



Definition 1 (Bounded Levenshtein-CFL reachability). *Given a CFL, ℓ, and an invalid string, $\underset{\sim}{\sigma} : \bar\ell$, find every valid string reachable within d edits of $\underset{\sim}{\sigma}$, i.e., if Δ_L is the Levenshtein metric and $\mathcal{L}\big(L(\underset{\sim}{\sigma}, d)\big) = \{\sigma' \mid \Delta_L(\underset{\sim}{\sigma}, \sigma') \leq d\}$ is the Levenshtein d-ball, we seek to find $\ell_\cap = \mathcal{L}\big(L(\underset{\sim}{\sigma}, d)\big) \cap \ell$.*

Moreover, we want a procedure that surfaces *both* natural and valid repairs:

Definition 2 (Ranked repair). *Given a finite language $\ell_\cap = \mathcal{L}\big(L(\underset{\sim}{\sigma}, d)\big) \cap \ell$ and LM, $P_\theta : \Sigma^* \to \mathbb{R}$, find the top-$k$ maximum probability repairs, i.e.,*

$$R(\ell_\cap, P_\theta) : 2^{\Sigma^+} \times (\Sigma^+ \to \mathbb{R}) \to (\Sigma^+)^{\leq k} = \underset{\boldsymbol{\sigma} \subseteq \ell_\cap, |\boldsymbol{\sigma}| \leq k}{\arg\max} \sum_{\sigma \in \boldsymbol{\sigma}} P_\theta(\sigma)$$

Evaluation

We will compare four different methods for constrained decoding from $\ell_\cap$:

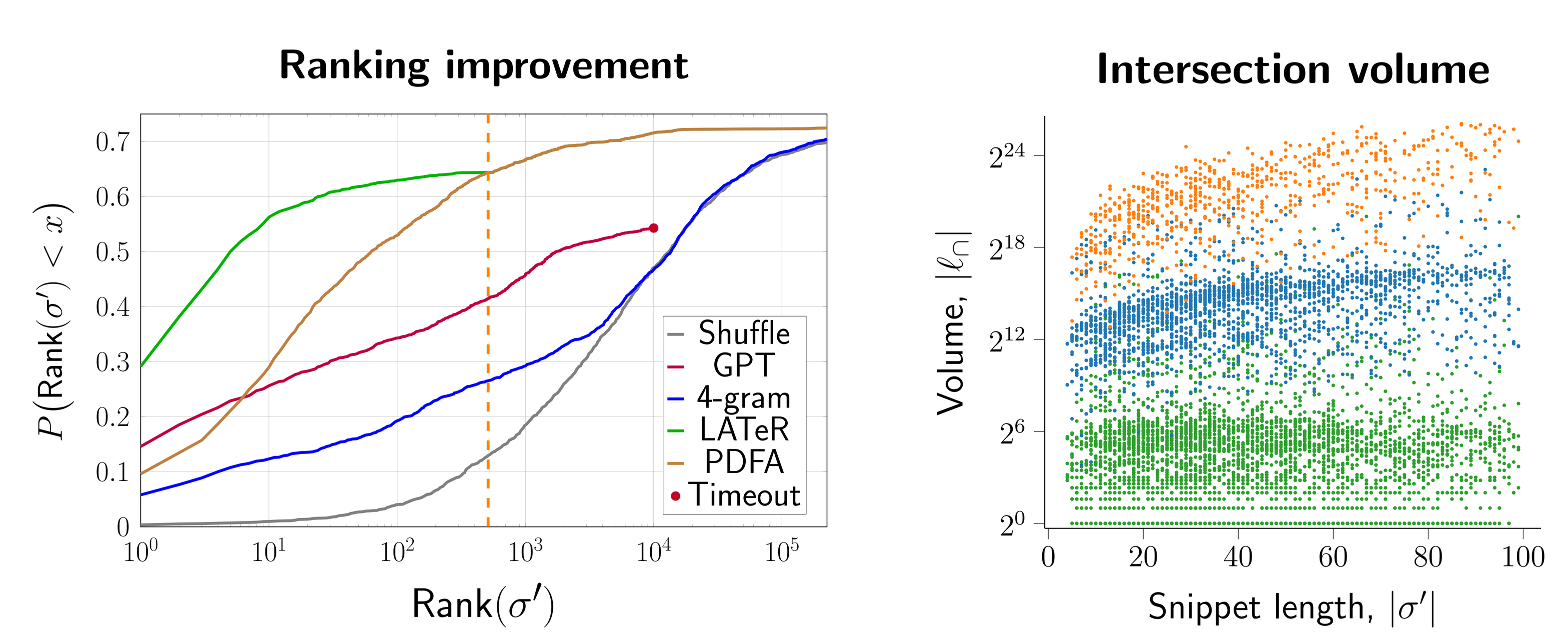
1. A **4-gram** model with Laplace smoothing trained on $\sigma \in \mathcal{L}(G)$
2. A **GPT**-style model trained on $\underset{\sim}{\sigma}\sigma'$ with LTR σ' masking
3. A **PDFA** distilled from P_θ overapproximating $\mathcal{L}(G)$
4. A **LATeR** (PDFA decoding + top-512 reranker)

Let $\mathsf{E}_\theta : \Sigma^+ \to \mathbb{R}^p$ be a transformer encoder and $E'_\theta : \Sigma^+ \times \mathbb{Z}_4^+ \to \mathbb{R}^p$ be an encoder which takes in a Levenshtein alignment $(\mathsf{LA} : \Sigma^+ \times \Sigma^+ \to \mathbb{Z}_4^+)$ for each $\langle \underset{\sim}{\sigma}, \sigma\rangle$ pair. Now add an MLP and output a scalar relevance score:

$$\Delta_\theta(\underset{\sim}{\sigma}, \sigma) = \mathsf{MLP}_\theta\Big(\mathsf{E}_\theta(\underset{\sim}{\sigma}), \mathsf{E}'_\theta\big(\sigma, \mathsf{LA}(\underset{\sim}{\sigma}, \sigma)\big)\Big)$$

We call this model a Levenshtein-Aligned Transformer Reranker (LATeR) and train on $\langle \underset{\sim}{\sigma}, \sigma'\rangle$ with PDFA-sampled confounders and a listwise CE loss.

Results



Depicted is the cumulative rank and intersection volume across the test set.

Conclusion

- Single-threaded PDFA sampling exhibits $10^3 - 10^5\times$ higher throughput than constrained LLM decoding, often saturating $\ell_\cap$ in under 100ms.
- The size-k subset maximizing P_θ over $\ell_\cap$ can far outweigh the mass from an LLM-guided subsampling. We call this the **decoder-retriever gap**.
- Hence, exhaustive ranking with a low-fidelity but fast P_θ approximation is preferable to decoding a highly calibrated but low-throughput LLM.
- Is there a succinct construction (possibly with a very different topology) that soundly overapproximates $\mathcal{L}(G)$ and more faithfully reconstructs P_θ?